给定一个可对角化矩阵 H，求其模最大的本征值 E_0 及相应 $|E_0\rangle$

① Power interation

满足 $\langle E_0|V_0\rangle \neq 0$，将 H 不断作用在随机初态 $|V_0\rangle$ 上，认为最后的迭代结果为 $|E_0\rangle$

Pf: $|V_n\rangle \equiv H^n|V_0\rangle = H^n \sum_i |E_i\rangle\langle E_i|V_0\rangle$

$$= \sum_i E_i^n \langle E_i|V_0\rangle |E_i\rangle$$

as $n \nearrow$，$|E_0\rangle$ 在 $|V_n\rangle$ 中比重的增长快于其他本征态

n 足够大时，$|V_n\rangle$ 近似认为是 $|E_0\rangle$

② above 的迭代序列 $\{|V_0\rangle, |V_1\rangle, \cdots, |V_{n-1}\rangle\}$

$= \{|V_0\rangle, H|V_0\rangle, \cdots, H^{n-1}|V_0\rangle\}$ 称为 Krylov 序列

Krylov 序列张成的子空间称为 Krylov 空间.

具体地，由 $H, |V_0\rangle$ 张成的 n 阶 Krylov space 记为

$$K_n(H, |V_0\rangle) = \mathrm{span}\{|V_0\rangle, H|V_0\rangle, \cdots, H^{n-1}|V_0\rangle\}$$

* 可以找一种方法，在 Krylov space 中对角化 H，然后将子空间的基态作为下一次迭代中张成 Krylov space 的初态，加快收敛速度.

③ 为了在 Krylov space 中对角化 H，首先找一种 Krylov space 的正交基. $\{|\varphi_i\rangle\}$

$$|\varphi_i\rangle = \frac{1}{b_{i-1}}\left(H|\varphi_{i-1}\rangle - a_{i-1}|\varphi_{i-1}\rangle - b_{i-2}|\varphi_{i-2}\rangle\right)$$

这一过程称为 Krylov 正交化，其中

$$a_i = \langle \varphi_i | H | \varphi_i \rangle \quad \in \mathbb{R}$$

$$b_i = \langle \varphi_{i+1} | \varphi_{i+1} \rangle = \langle \varphi_i | H | \varphi_{i+1} \rangle \quad \in \mathbb{R}$$

initial condition: $|\varphi_0\rangle = \dfrac{|v_0\rangle}{\langle v_0 | v_0 \rangle}$, $|\varphi_{-1}\rangle = 0$, $b_{-1} = 0$

* $\{|\varphi_i\rangle\}$ 正交性:

1° $\langle \varphi_0 | \varphi_1 \rangle = \langle \varphi_0 | \left(\frac{1}{b_0} \left(H|\varphi_0\rangle - a_0 |\varphi_0\rangle - b_{-1} |\varphi_{-1}\rangle \right) \right)$

$$= \frac{1}{b_0} \left(\langle \varphi_0 | H | \varphi_0 \rangle - a_0 \langle \varphi_0 | \varphi_0 \rangle - b_{-1} \langle \varphi_0 | \varphi_{-1} \rangle \right)$$

$$= \frac{1}{b_0} \left(a_0 - a_0 - 0 \right) = 0$$

假设 $\langle \varphi_{i-1} | \varphi_i \rangle = 0$，则

$$\langle \varphi_i | \varphi_{i+1} \rangle = \frac{1}{b_i} \left(\langle \varphi_i | H \varphi_i \rangle - a_i \langle \varphi_i | \varphi_i \rangle - b_{i-1} \langle \varphi_i | \varphi_{i-1} \rangle \right)$$

$$= \frac{1}{b_i} \left(a_i - a_i \times 1 - b_{i-1} \times 0 \right) = 0$$

$$\Longrightarrow \boxed{\langle \varphi_{i-1} | \varphi_i \rangle = 0 \quad \forall i.}$$

2° Note that $\langle \varphi_0 | \varphi_2 \rangle = \frac{1}{b_1} \left(\langle \varphi_0 | H | \varphi_1 \rangle - a_1 \langle \varphi_0 | \varphi_1 \rangle - b_0 \langle \varphi_0 | \varphi_0 \rangle \right)$

$$= \frac{1}{b_1} \left(\left(b_0 \langle \varphi_1 | + a_0 \langle \varphi_0 | + b_{-1} \langle \varphi_{-1} | \right) | \varphi_1 \rangle - a_1 \cdot 0 - b_0 \cdot 1 \right) = 0$$

假设 $\langle \varphi_{i-2} | \varphi_i \rangle = 0$，则

$$\langle \varphi_{i-1} | \varphi_{i+1} \rangle = \frac{1}{b_i} \left(\langle \varphi_{i-1} | H | \varphi_i \rangle - a_i \langle \varphi_{i-1} | \varphi_i \rangle - b_{i-1} \langle \varphi_{i-1} | \varphi_{i-1} \rangle \right)$$

$$= \frac{1}{b_i} \left(\left(b_{i-1} \langle \varphi_i | + a_{i-1} \langle \varphi_{i-1} | + b_{i-2} \langle \varphi_{i-2} | \right) | \varphi_i \rangle - a_i \cdot 0 - b_{i-1} \cdot 1 \right)$$

$$= \frac{1}{b_i} \left(b_{i-1} \cdot 1 + a_{i-1} \cdot 0 + b_{i-2} \cdot 0 - a_i \cdot 0 - b_{i-1} \cdot 1 \right) = 0$$

$$\Rightarrow \boxed{\langle \varphi_{i-2} | \varphi_i \rangle = 0 \ , \ \forall i}$$

$$\Longrightarrow \boxed{\langle \varphi_i | \varphi_j \rangle = \delta_{ij}}$$

$$\begin{cases} \langle \varphi_i | H | \varphi_i \rangle = a_i \\ \langle \varphi_i | H | \varphi_{i+1} \rangle = \left(b_i \langle \varphi_{i+1} | + a_i \langle \varphi_i | + b_{i-1} \langle \varphi_{i-1} | \right) | \varphi_{i+1} \rangle = b_i \\ \langle \varphi_i | H | \varphi_{i-1} \rangle = \left(b_i \langle \varphi_{i+1} | + a_i \langle \varphi_i | + b_{i-1} \langle \varphi_{i-1} | \right) | \varphi_{i-1} \rangle = b_{i-1} \\ \langle \varphi_i | H | \varphi_{i \pm k} \rangle \overset{k>1}{=\!=} 0 \end{cases}$$

$\Longrightarrow$ 存 $\{ |\varphi_i\rangle \}$ 下，

$$H = \begin{pmatrix} a_0 & b_0 & & & 0 \\ b_0 & a_1 & b_1 & & \\ & b_1 & \ddots & \ddots & \\ & & \ddots & \ddots & b_{n-2} \\ 0 & & & b_{n-2} & a_{n-1} \end{pmatrix}$$

$$|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle$$

$$e^{-iHt} = \sum_{n=0}^{\infty} \frac{(-iHt)^n}{n!} = 1 - iHt + \frac{-H^2}{2}t^2 + \cdots$$

$\{H^n|\psi(0)\rangle \mid n \in \mathbb{Z}\}$ basis

$$H|K_n\rangle = a_n|K_n\rangle + b_n|K_{n-1}\rangle + b_{n+1}|K_{n+1}\rangle$$

$$\Rightarrow |\psi(t)\rangle = \sum_n \psi_n(t)|K_n\rangle \qquad \left(|\psi(t)\rangle = \sum_n |K_n\rangle \underbrace{\langle K_n|\psi(t)\rangle}_{\psi_n(t)}\right)$$

where $\psi_n(t)$ are obtained by

$$i\partial_t \psi_n(t) = a_n\psi_n(t) + b_{n+1}\psi_{n+1}(t) + b_n\psi_{n-1}(t)$$

$$\langle K_n|\psi(t)\rangle = \langle K_n|e^{-iHt}|\psi(0)\rangle$$

$$\approx \langle K_n|1 - iHt|\psi(0)\rangle$$

$$= \langle K_n|\psi(0)\rangle - i\langle K_n|Ht|\psi(0)\rangle$$

$t \to 0$

Conservation of probability: $\sum_n |\psi_n(t)|^2 = 1$

$\psi_0(t) = \langle K_0|\psi(t)\rangle = \langle\psi(0)|\psi(t)\rangle \equiv S(t)$ is often called the return amplitude.

<u>Def</u> complexity

$$C(t) = \sum_n n|\psi_n(t)|^2$$

which is minimized by the above choice of Krylov basis.

q - deformed $SU(2)$ algebra

$$[x]_q := \frac{q^x - q^{-x}}{q - q^{-1}} \quad , \quad \lim_{q \to 1} [x]_q = x$$

q can be real or complex
$\qquad e^{\tau} \qquad e^{i\tau}$

We take $p > 0 \in \mathbb{R}$

e.g. $[1]_q = 1$, $[2]_q = q + q^{-1}$, $[3]_q = 1 + q^2 + q^{-2}$

Note that $[n]_q = U_{n-1}(x)$, $x = \frac{q + q^{-1}}{2}$

$\hookrightarrow$ Chebyshev polynomials

$*$ $[x]_q = [x]_{\frac{1}{q}}$ $\Rightarrow$ we take $0 < q \leq 1$

The q-deformed $SU(2)$, denoted as $SU_q(2)$, is generated by three generators J_0, J_+, J_- with:

$$[J_0, J_\pm] = \pm J_\pm \, , \quad [J_+, J_-] = [2J_0]_q$$

$$\begin{cases} [J_+, J_-] = \dfrac{1}{\sinh \tau} \displaystyle\sum_{n=0}^{\infty} \frac{(2\tau J_0)^{2n+1}}{(2n+1)!} & q = e^{\tau} \\[2ex] [J_+, J_-] = \dfrac{1}{\sin \tau} \displaystyle\sum_{n=0}^{\infty} (-1)^n \frac{(2\tau J_0)^{2n+1}}{(2n+1)!} & q = e^{i\tau} \end{cases}$$

$\Rightarrow$ $SU_q(2)$ is a non-linear generalization of the standard $SU(2)$. $\lim_{q \to 1} SU_q(2) = SU(2)$

natural basis : $|j, n\rangle \qquad -j \le n \le j$

$$|j, -j+n\rangle = \sqrt{\frac{[\Gamma(2j-n+1)]_q}{[n]_q!\,[\Gamma(2j+1)]_q}}\; J_+^n |j, -j\rangle$$

where $n = 0, 1, \cdots, 2j$

$$\begin{cases} J_0 |j, -j+n\rangle = (-j+n)\,|j, -j+n\rangle \\ J_+ |j, -j+n\rangle = \sqrt{[n+1]_q\,[2j-n]_q}\;|j, -j+n+1\rangle \\ J_- |j, -j+n\rangle = \sqrt{[n]_q\,[2j-n+1]_q}\;|j, -j+n-1\rangle \end{cases}$$

$$\alpha_n^{(q)} = \eta_0\,(-j+n) + \delta$$

$$b_n^{(q)} = \alpha\sqrt{[n]_q\,[2j-n+1]_q} = \alpha\sqrt{U_{n-1}(x)\,U_{2j-n}(x)}$$

$$= \alpha\left[\sum_{k=0}^{n-1} U_{2j-2n+2k+1}(x)\right]^{\frac{1}{2}}$$